

Module-4 (Assignment Day 3)

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.



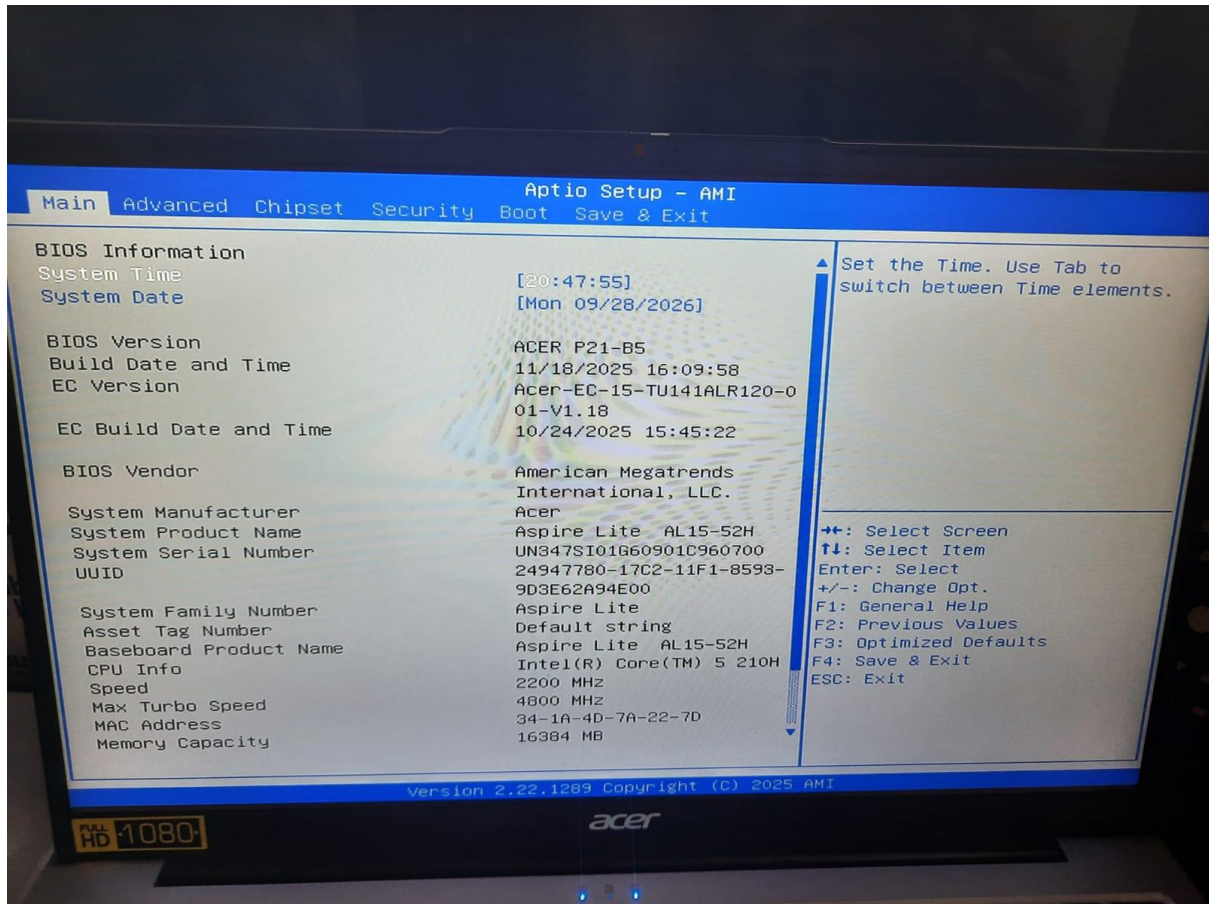
No.	BIOS/POST Error Message	Hardware / Configuration Issue
1	CMOS Checksum Error	CMOS battery may be weak or BIOS settings may be corrupted.
2	Keyboard Error / No Keyboard Detected	Keyboard may be disconnected, faulty, or USB port may have a problem.
3	Memory Error / RAM Error	RAM may be loose, faulty, or incompatible.
4	CPU Fan Error	CPU fan may not be connected, may have failed, or fan speed may be too low.
5	No Boot Device Found	HDD/SSD may not be detected, disconnected, or the boot order may be incorrect.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.



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3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.



CMOS Checksum Error – Defaults Loaded

Meaning:

The message “**CMOS checksum error – Defaults loaded**” means that the BIOS/UEFI detected that the stored BIOS settings are incorrect, corrupted, or lost. The system therefore loaded the **default BIOS settings**.

How to resolve it in a real system:

1. Enter the BIOS/UEFI Setup when the PC starts (usually by pressing Delete, F2, or F10).
2. Check the date and time and correct them if necessary.
3. Load Optimized/Default Settings, then save and restart.
4. If the error appears again, shut down the PC and unplug the power.
5. Check or replace the CMOS battery (usually a CR2032 coin-cell battery).
6. After replacing it, enter BIOS again and set the correct date, time, boot order, and other required settings.
7. Save the settings and restart the computer.
8. If the error continues even with a new battery, check for BIOS/UEFI corruption or a motherboard fault.

Conclusion: The most common cause is a **weak or dead CMOS battery**, which causes BIOS settings to reset.